4 **(a)** State what is represented by an electric field line.

.....

..... [2]

(b) Two point charges P and Q are placed 0.120 m apart as shown in Fig. 4.1.

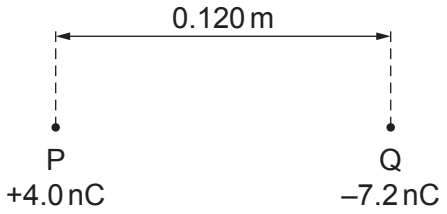


Fig. 4.1

(i) The charge of P is +4.0 nC and the charge of Q is –7.2 nC.

Determine the distance from P of the point on the line joining the two charges where the electric potential is zero.

distance = ... m [2]

(ii) State and explain, without calculation, whether the electric field strength is zero at the same point at which the electric potential is zero.

.....

.....

..... [1]

(iii) An electron is positioned at point X, equidistant from both P and Q, as shown in Fig. 4.2.

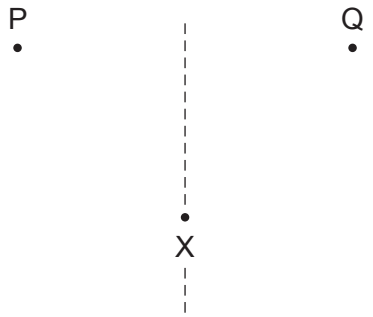


Fig. 4.2

On Fig. 4.2, draw an arrow to represent the direction of the resultant force acting on the electron. [1]

[Total: 6]

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